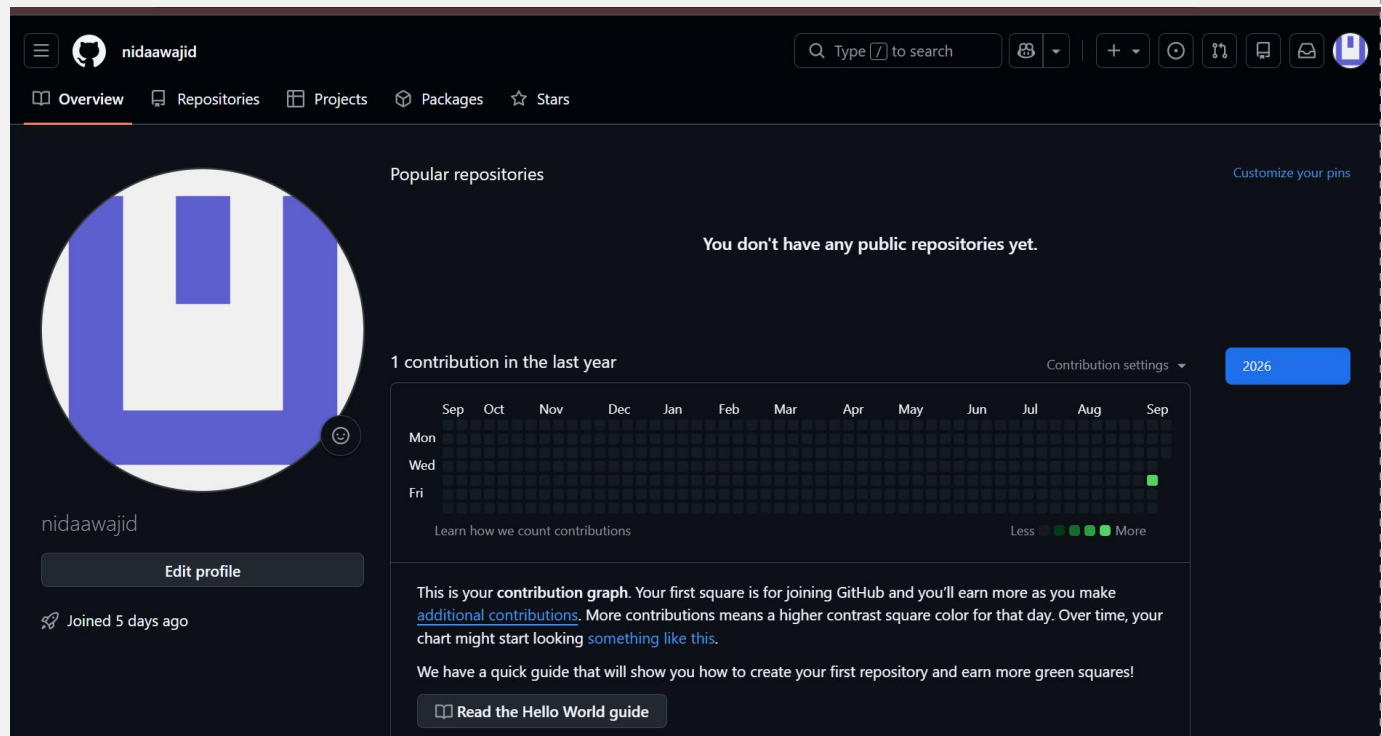

LAB NO 1

CLOUD COMPUTING

Lab Number & Lab Title:	Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation
Name:	Nida Wajid
Registration No:	FA24B1-SE-079
Department/Section:	Software Engineering / Section A
Course Code:	CC
GitHub Username:	nidaawajid

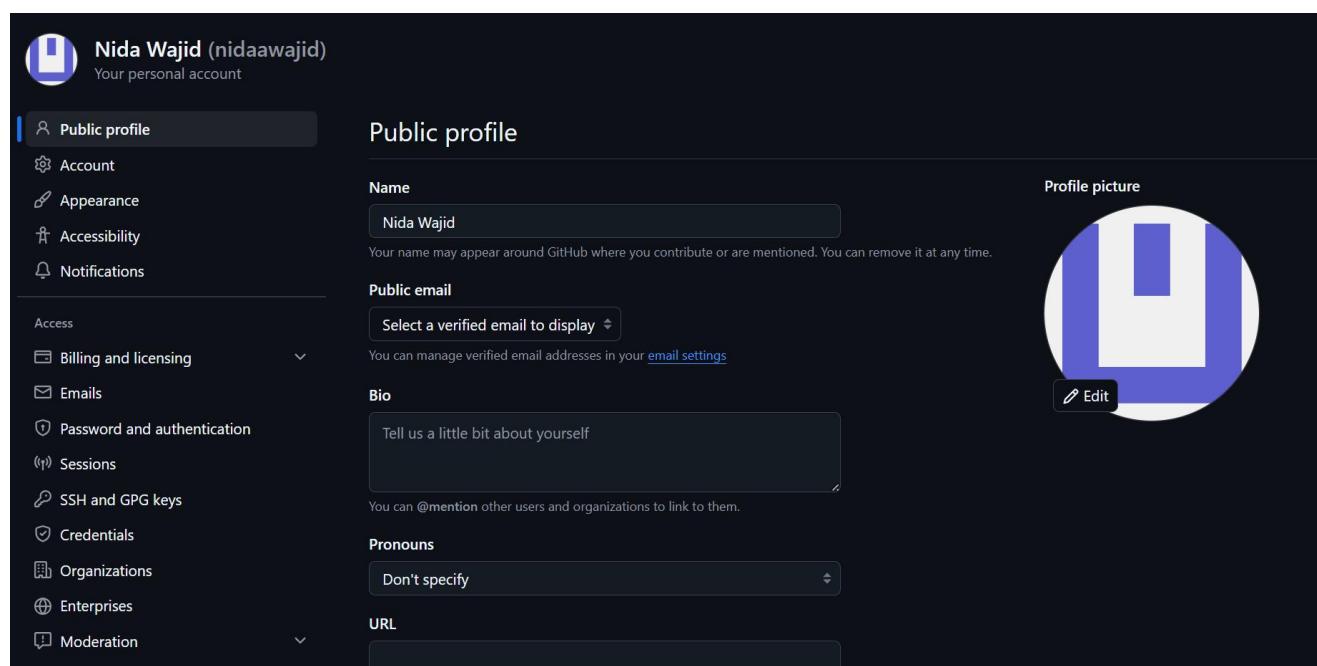
Task 1: Create and Configure Your GitHub Account

A personal GitHub account was created and the profile name was set to match my actual identity.



File name: github_profile.png

Caption:-Overview of personal GitHub profile showing username and account details

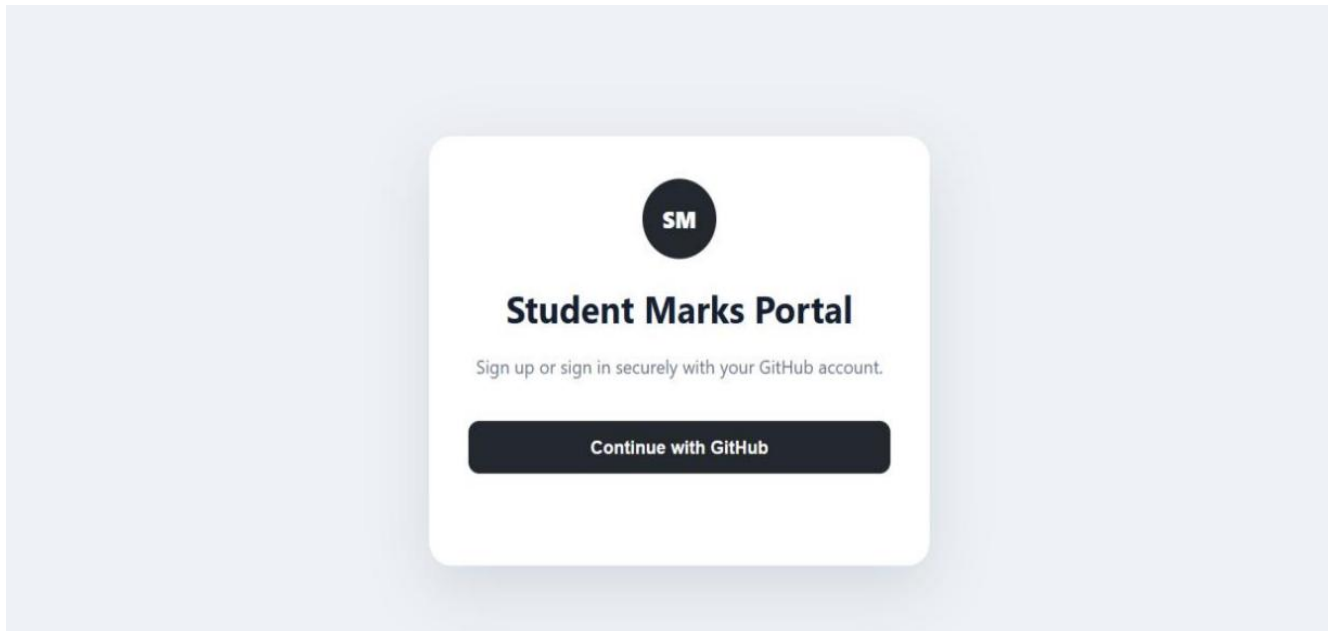


File name: github_actual_name.png

Caption:-Verification of complete actual name configured in GitHub Public Profile settings.

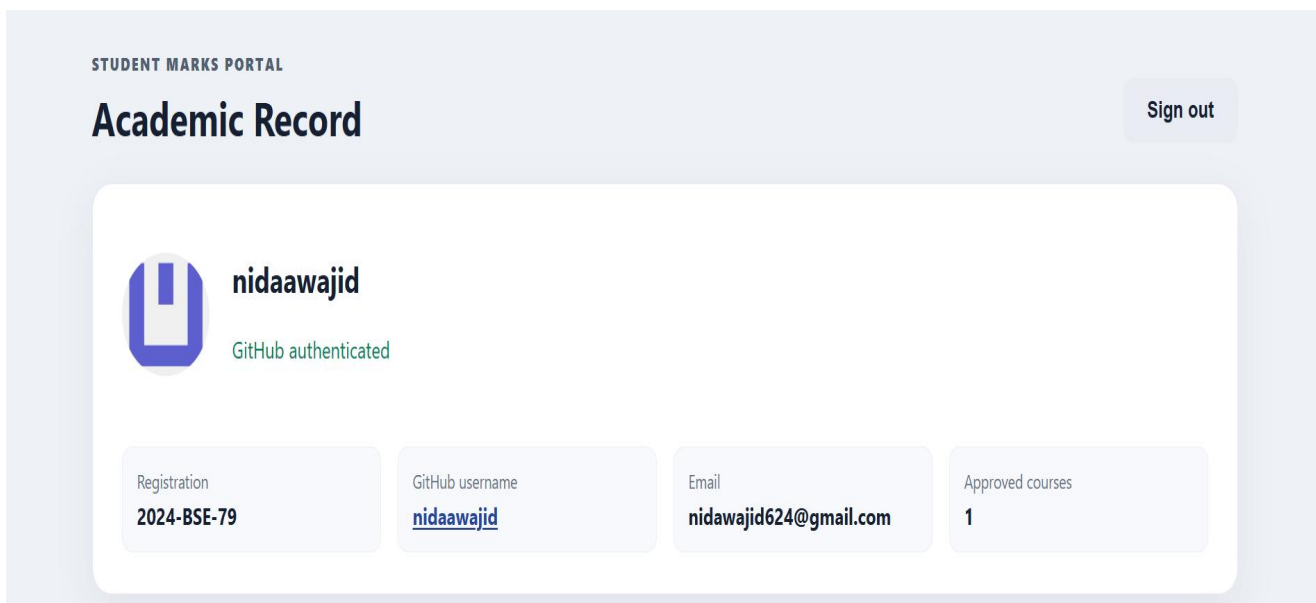
Task 2: Register in the Student Marks Portal

Enrollment was submitted on the Student Marks Portal using the GitHub account and correct registration details.



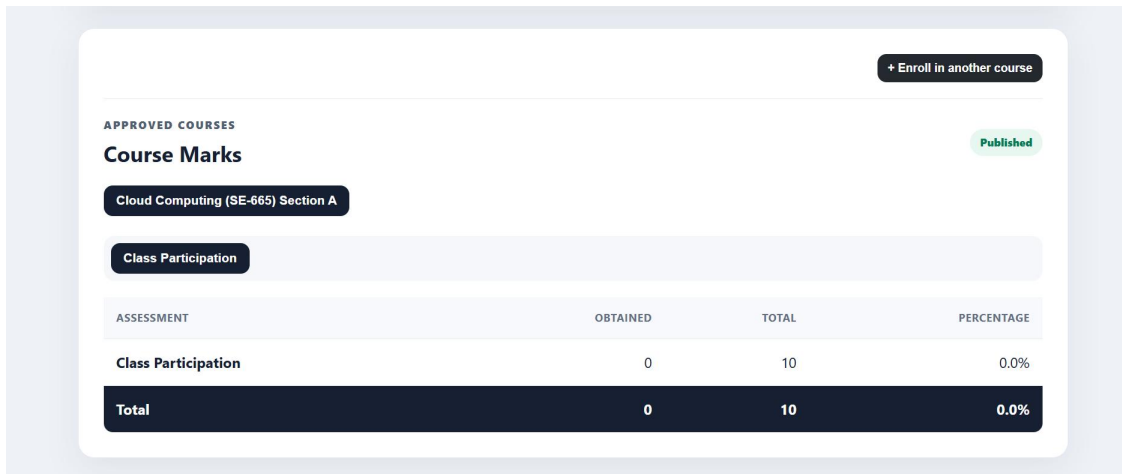
File name: portal_login_page.png

Caption:-Main login page of the Student Marks Portal.



File name: portal_github_profile.png

Caption:-GitHub profile integration inside the Student Marks Portal.



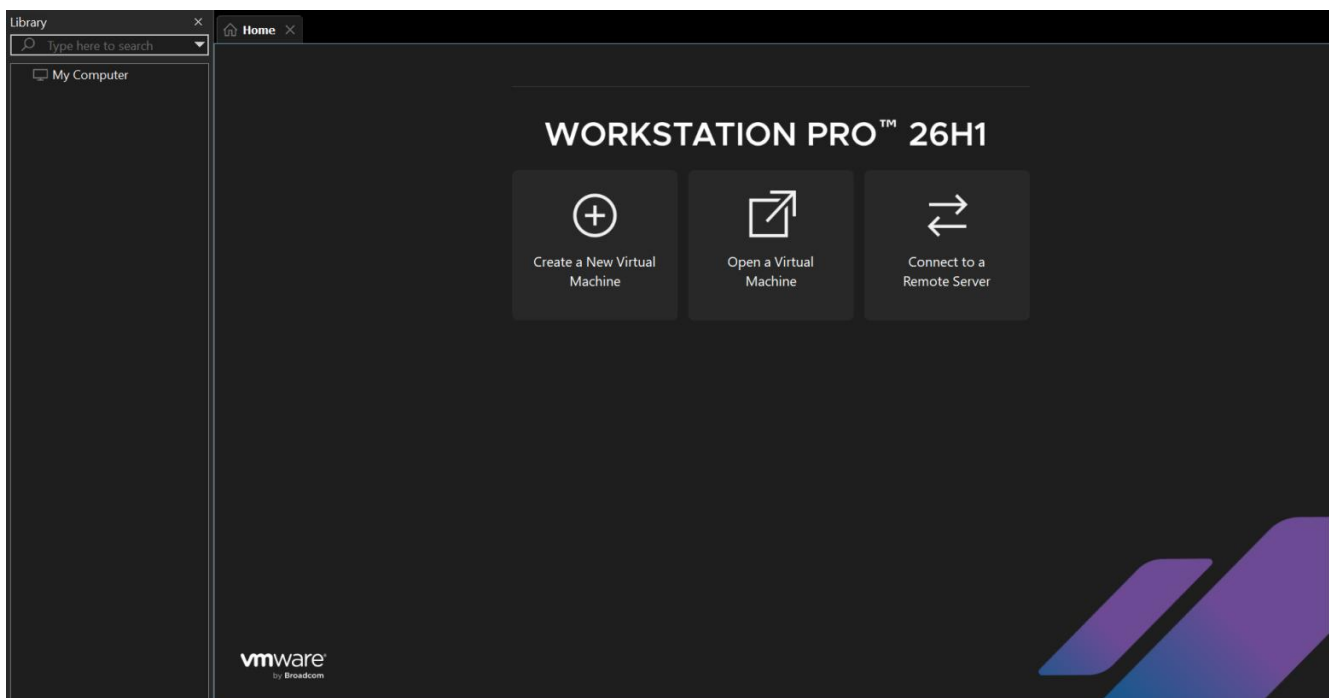
ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Class Participation	0	10	0.0%
Total	0	10	0.0%

File name: portal_enrollment_submitted.png

Caption:- Submitted student enrollment status showing registration number and course details.

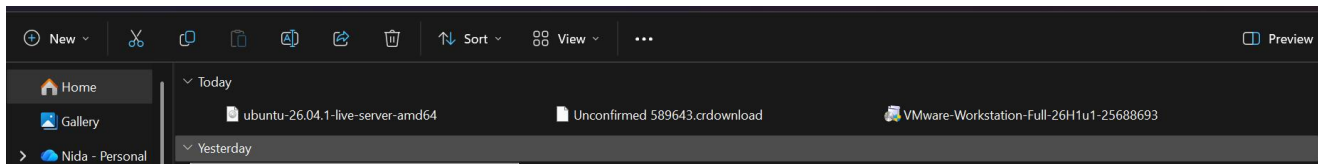
Task 3: Download VMware Workstation Pro and Ubuntu Server

VMware Workstation Pro was verified and the Ubuntu Server LTS ISO was downloaded with sufficient free storage confirmed.



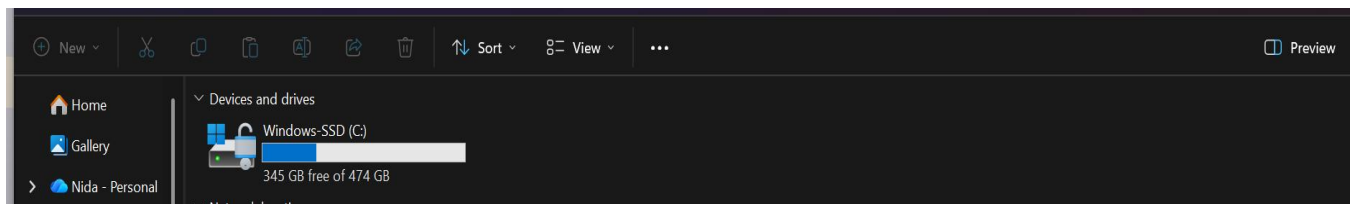
File name: vmware_workstation.png

Caption:-Verification of VMware Workstation Pro running on Windows.



File name: ubuntu_server_iso.png

Caption:-Downloaded Ubuntu Server LTS ISO image file

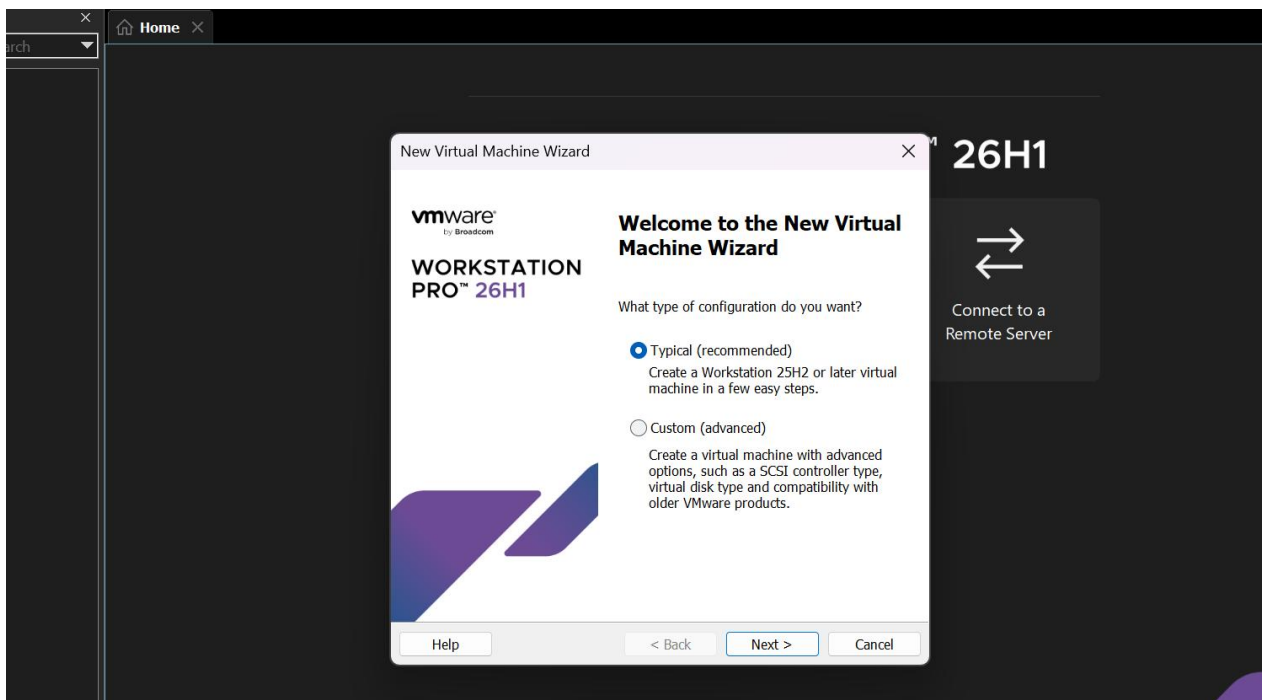


File name: available_storage.png

Caption:-Available system disk storage verification for virtual machine creation.

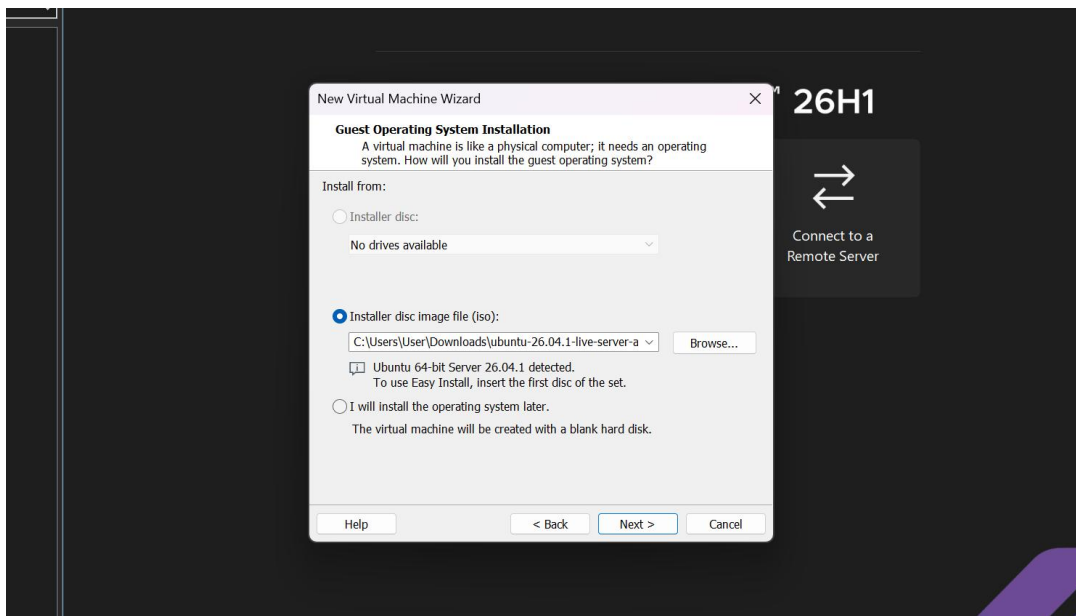
Task 4: Create the Ubuntu Server Virtual Machine

A new virtual machine was created in VMware Workstation Pro using the downloaded Ubuntu Server ISO.



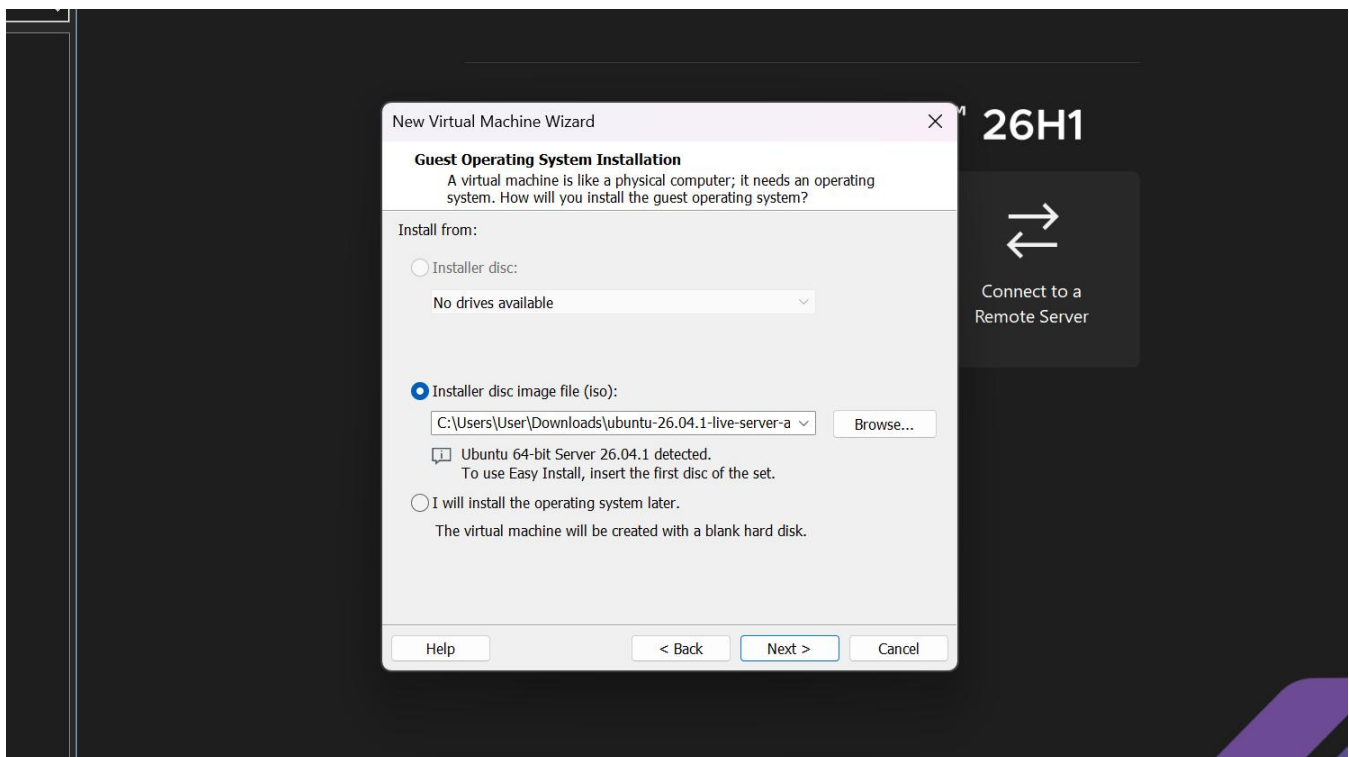
File name: vm_creation_wizard.png

Caption:-Initial step of VMware Virtual Machine creation wizard



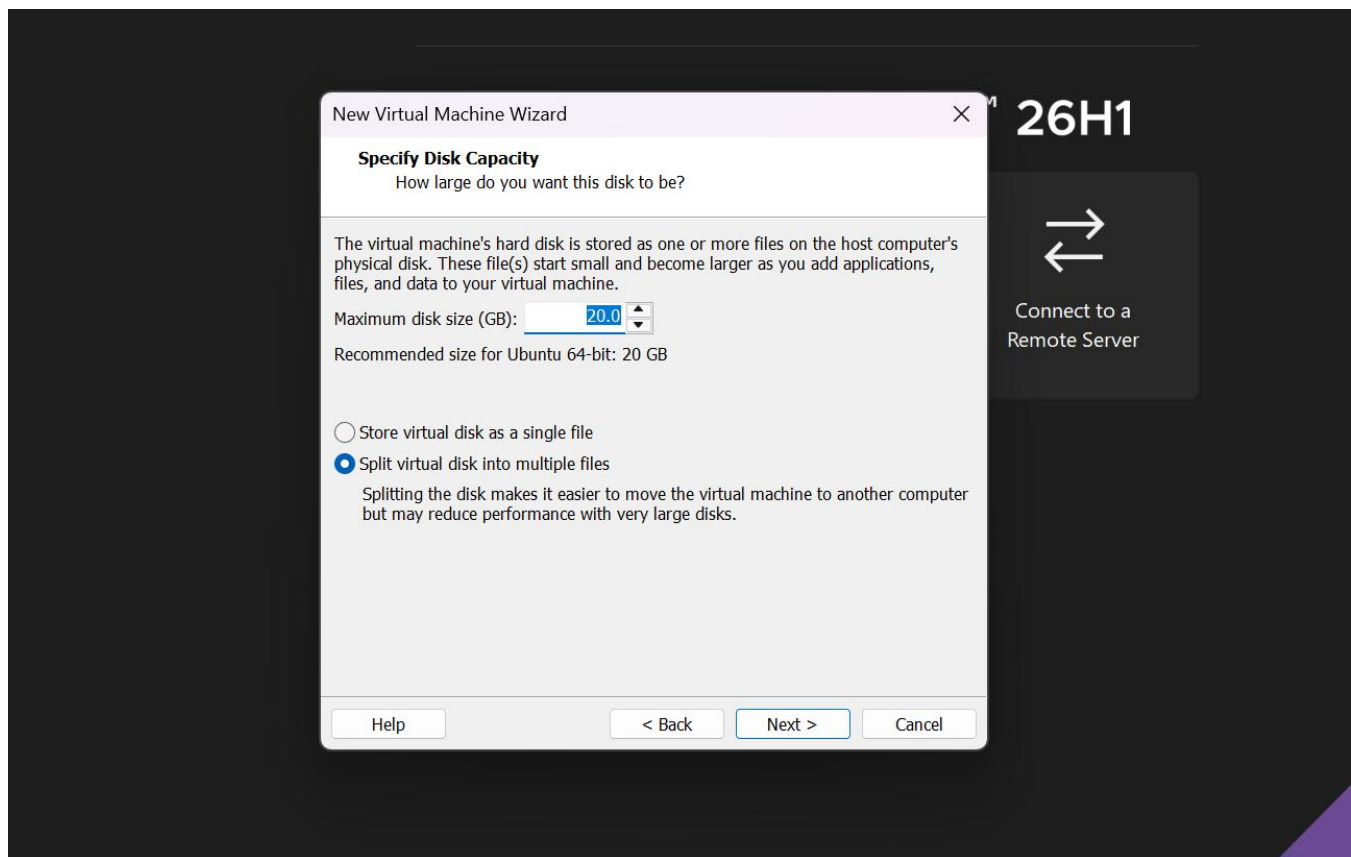
File name :vm_typical_configuration.png

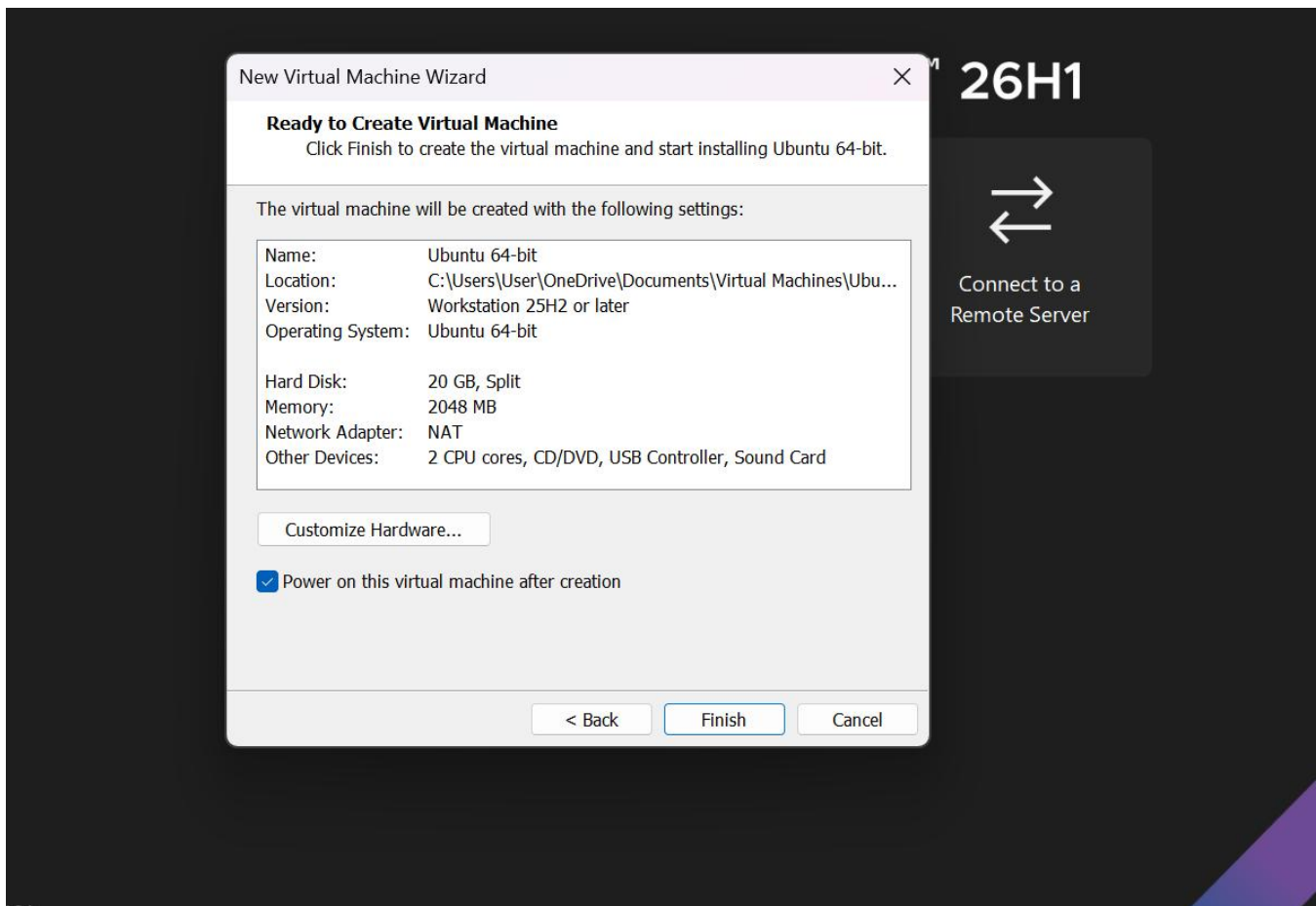
Caption:-Selecting Typical virtual machine configuration.



File name :vm_ubuntu_iso_selected.png

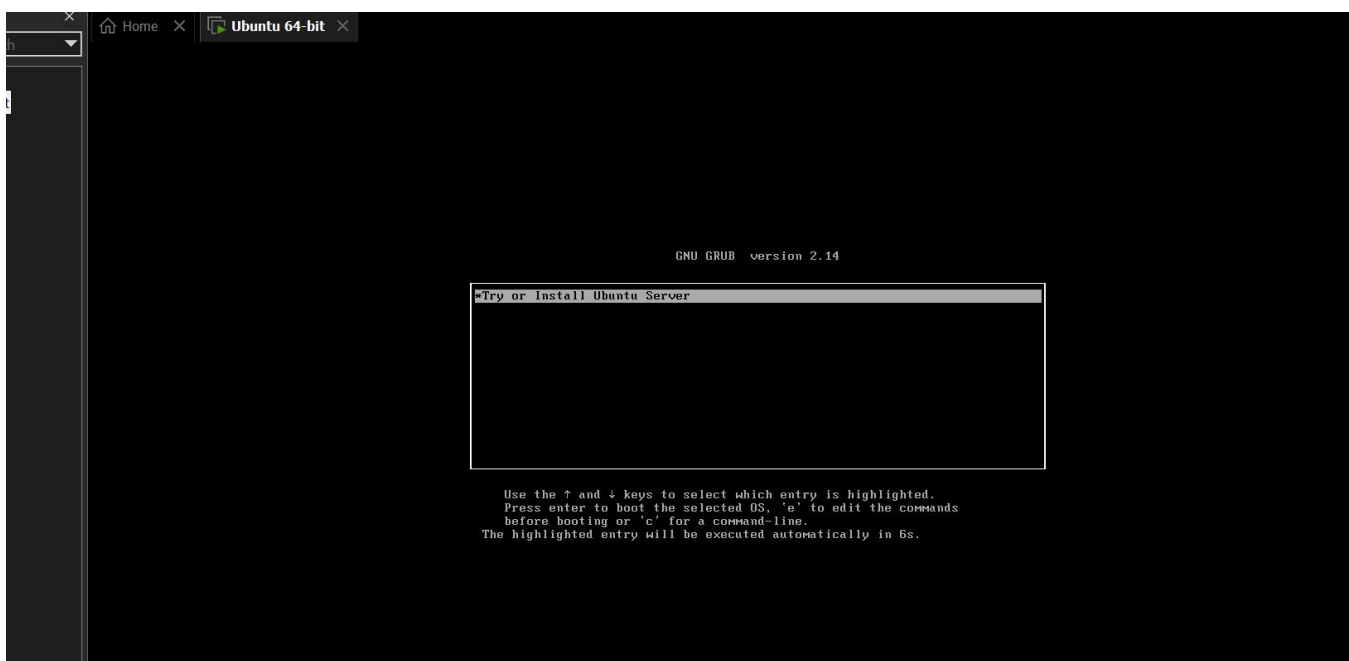
Caption:-Selecting the downloaded Ubuntu Server ISO image path





File Name: vm_configuration_summary.png

Caption:-Final hardware configuration summary for Ubuntu Server VM

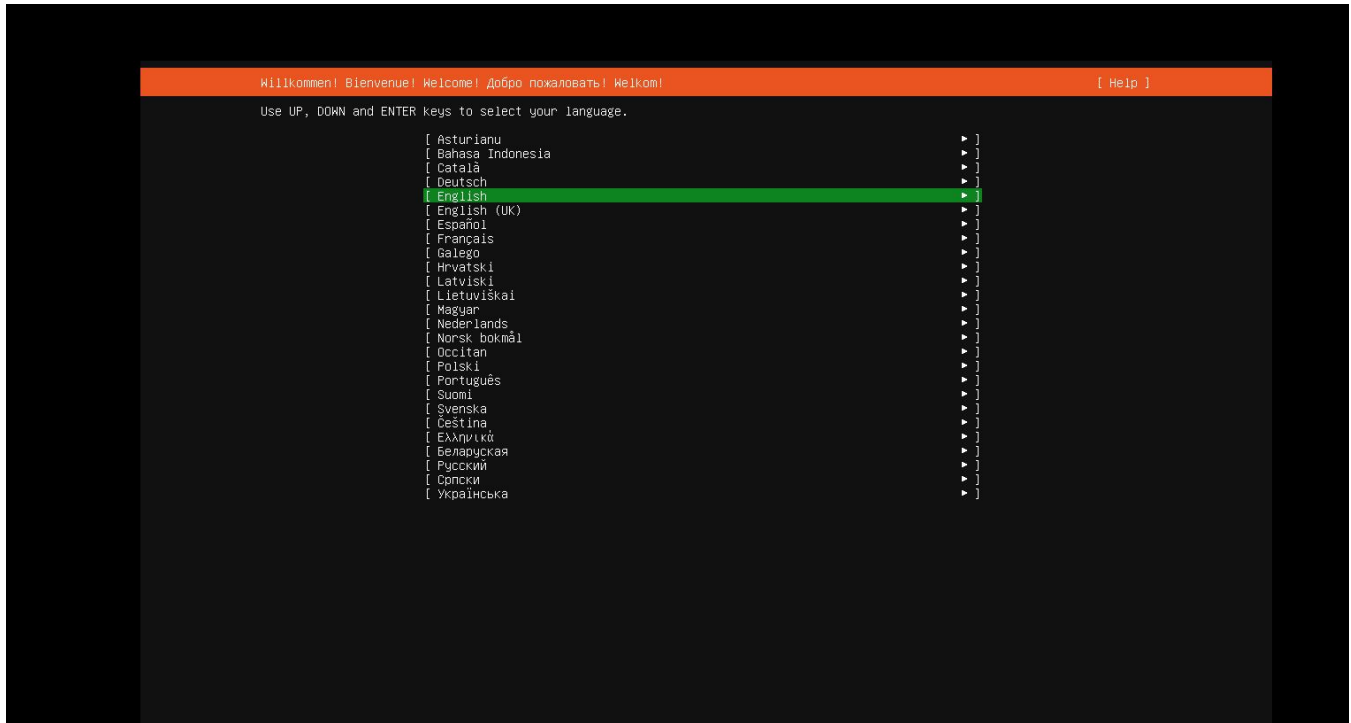


File Name:ubuntu_installer_boot.png

Caption:-Ubuntu Server Initial installer boot screen

Task 5: Choose the Installation Language

The installation language for the Ubuntu Server setup was selected.

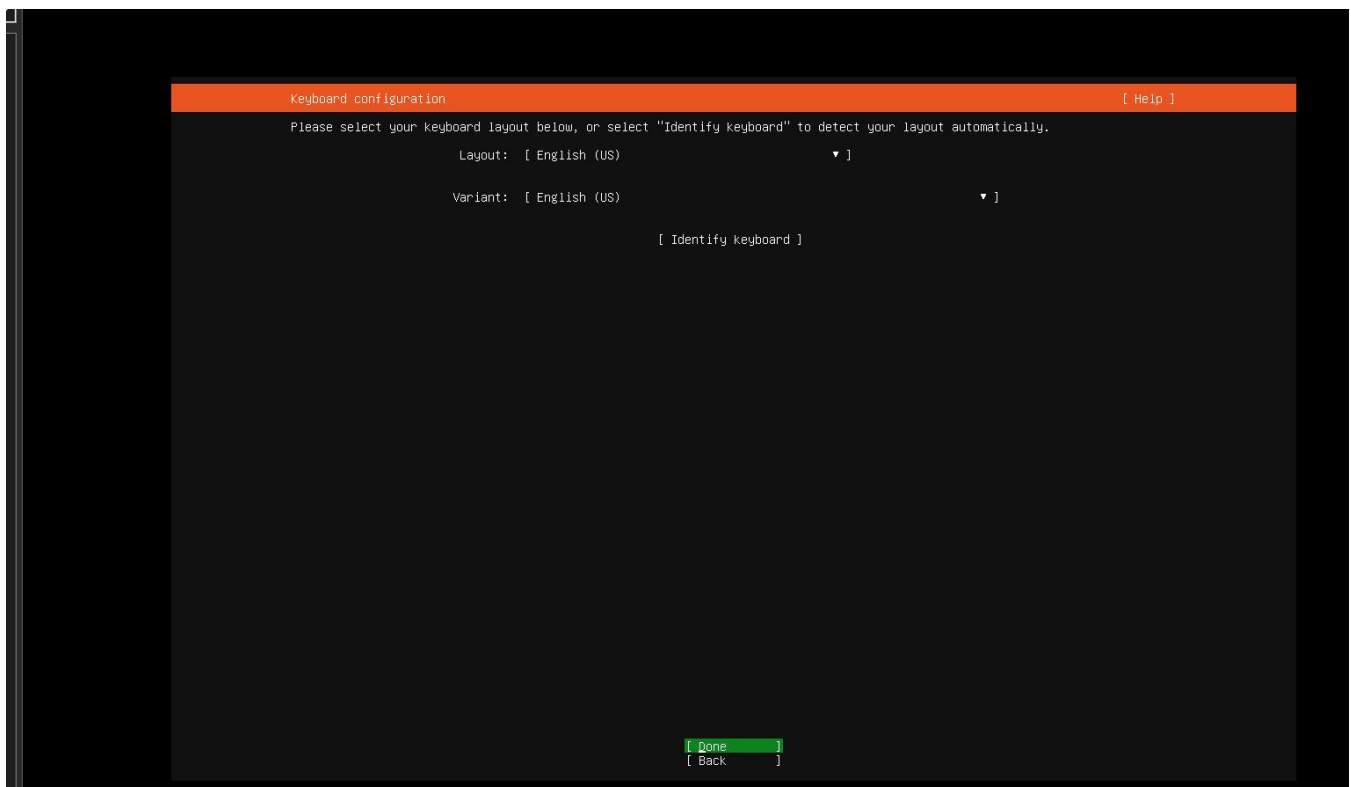


File name: ubuntu_language.png

Caption:- Selecting installer language in Ubuntu Server setup.

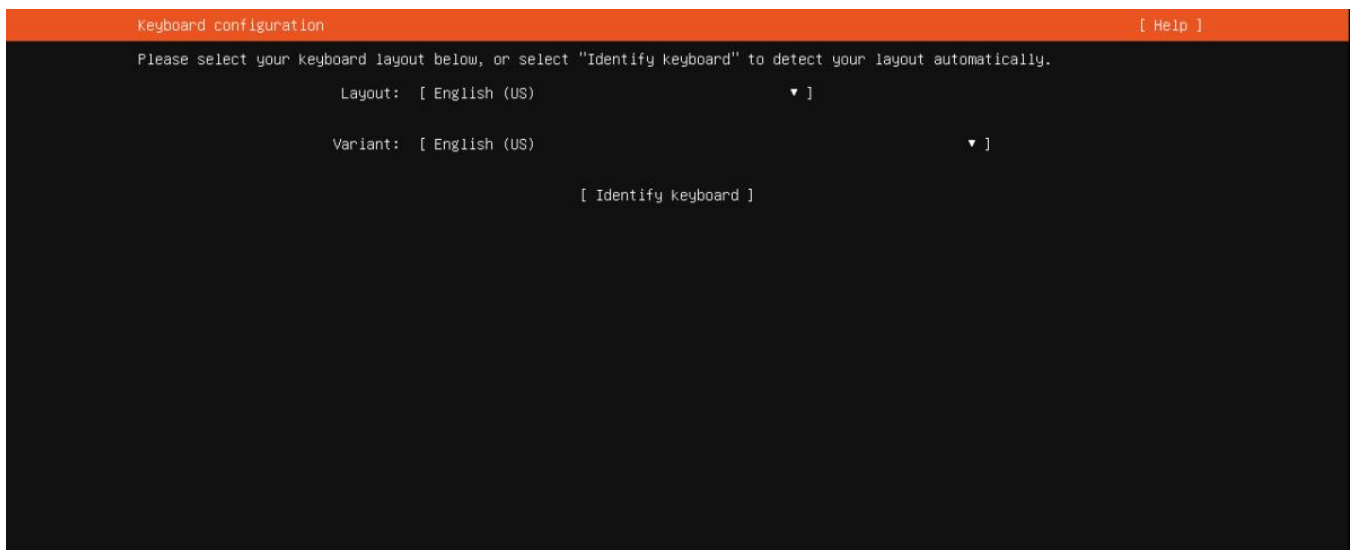
Task 6: Configure the Keyboard Layout

The keyboard layout and variant matching the physical keyboard were configured.



File name: ubuntu_keyboard_layout.png

Caption: Configuring keyboard layout.

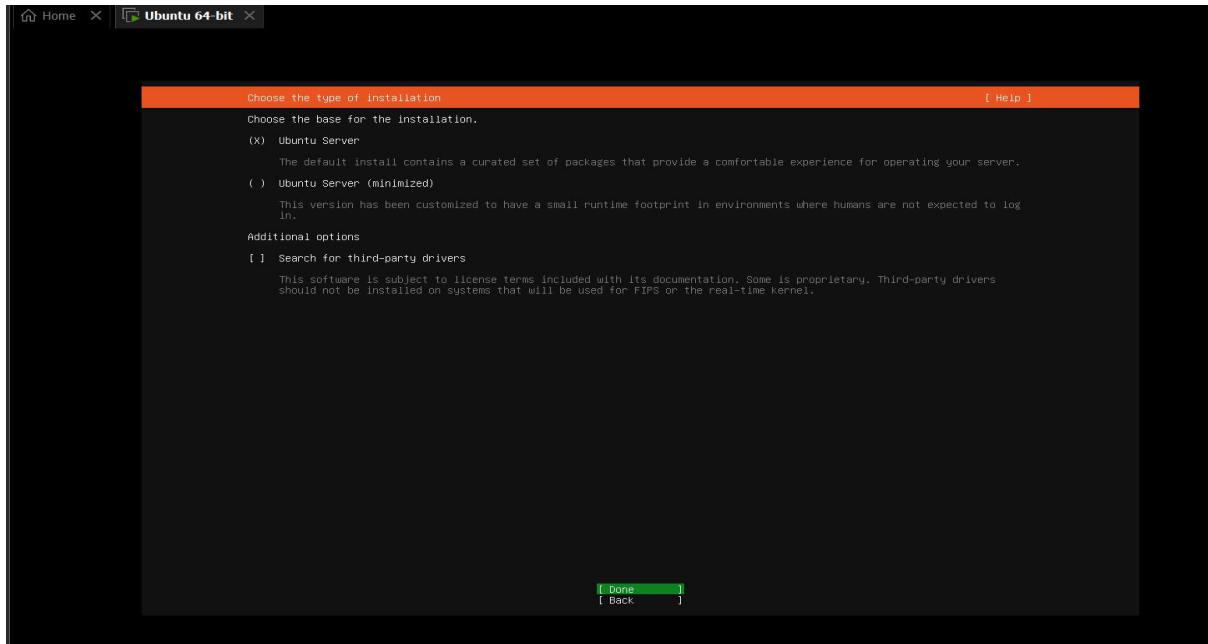


File Name: ubuntu_keyboard_variant.png

Caption:- Selecting keyboard variant settings.

Task 7: Choose the Ubuntu Server Installation Type

The standard Ubuntu Server installation type was selected to proceed with setup.

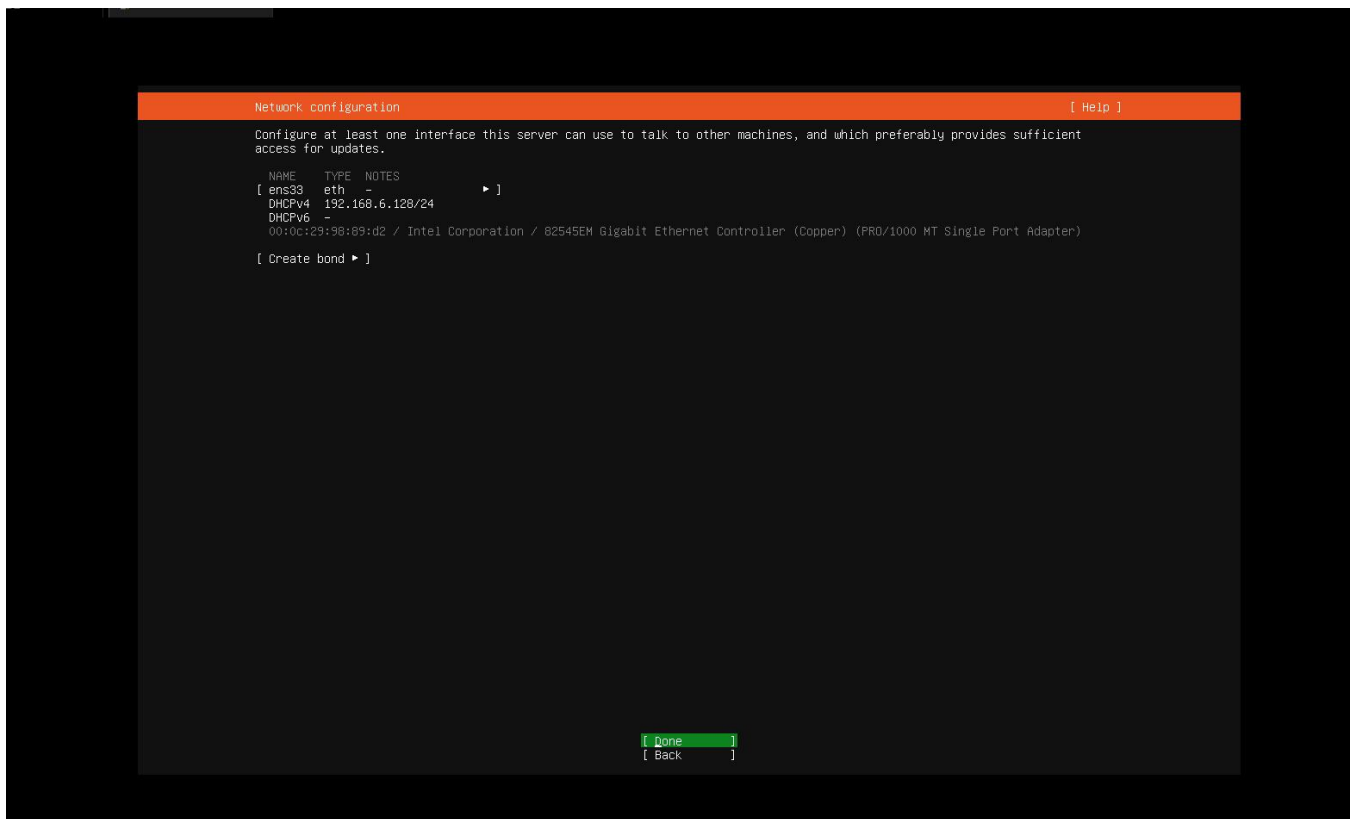


File name: ubuntu_installation_type.png

Caption:-Selecting default Ubuntu Server installation option

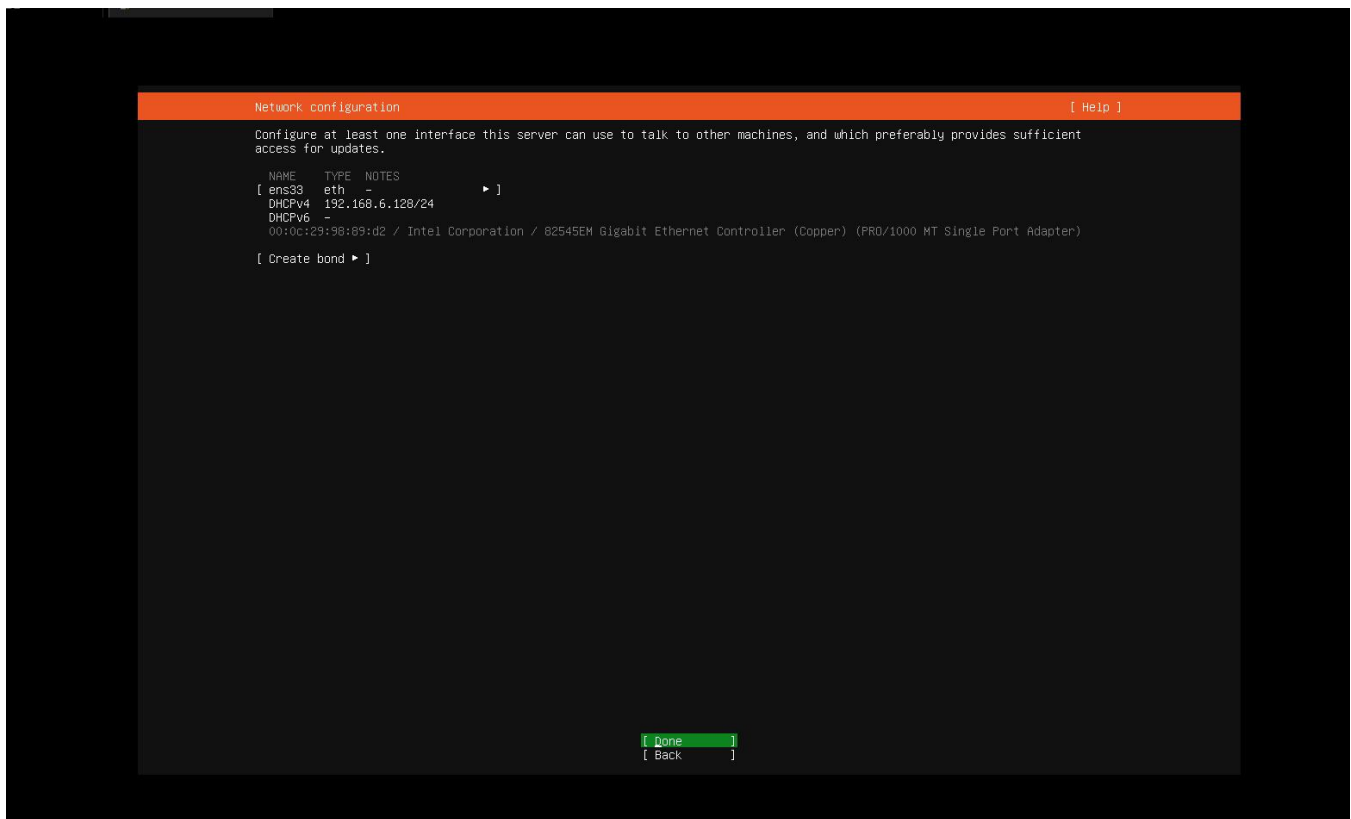
Task 8: Configure the Network

The detected network interface was reviewed and confirmed to have received an IP address via DHCP.



File name: ubuntu_network_interface.png

Caption:- Detection of the network interface.

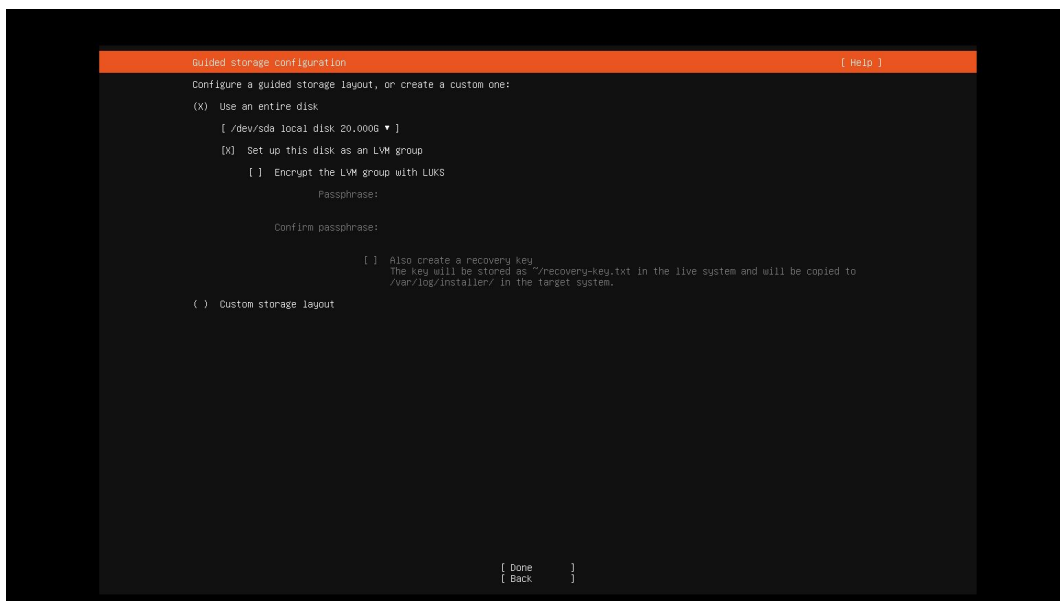


File name: ubuntu_installer_network.png

Caption:- Successful IPv4 DHCP address assignment on the network interface.

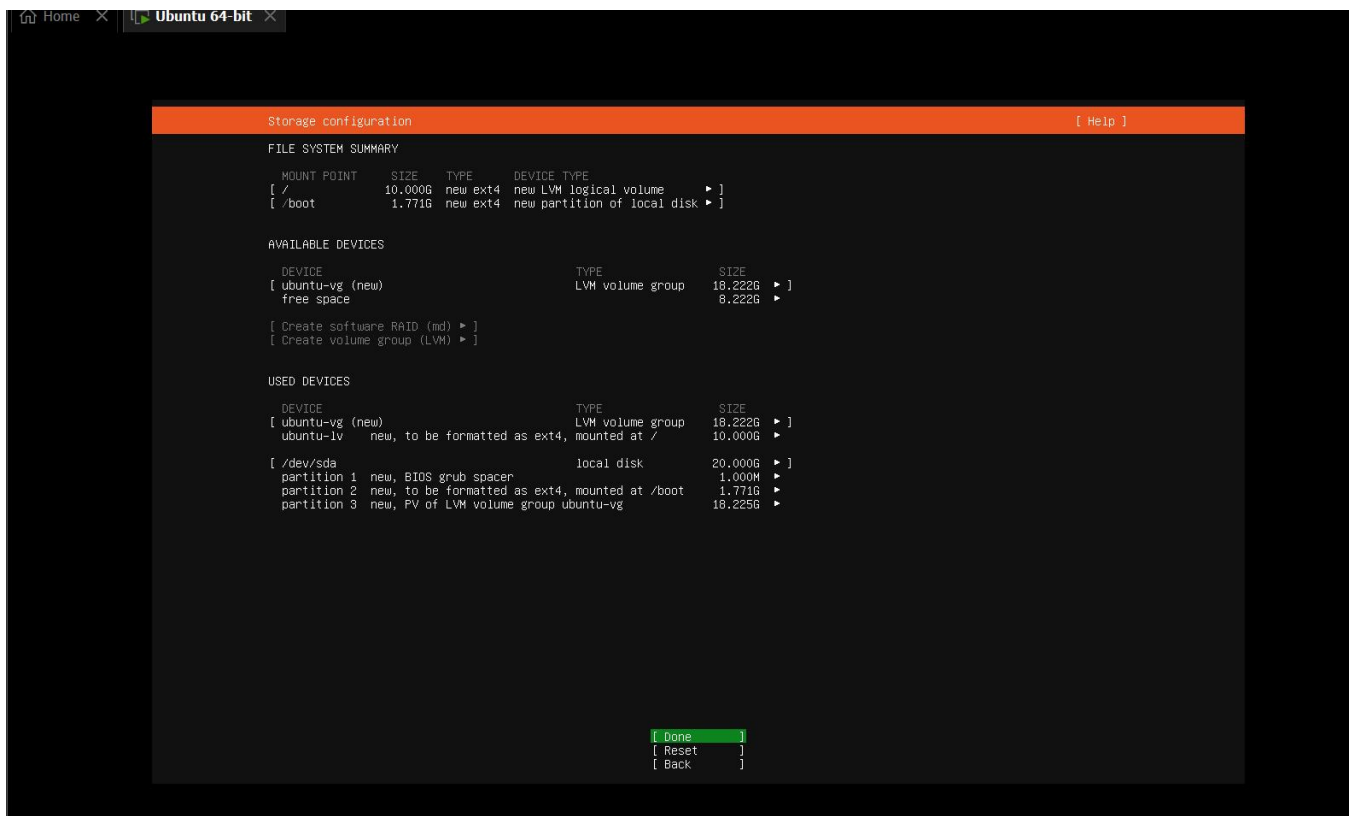
Task 9: Configure Storage

Guided storage configuration was selected to use the VMware virtual disk for installation.



File name: ubuntu_guided_storage.png

Caption:- Selecting guided storage setup on the VMware virtual disk.



File name: ubuntu_storage_configuration.png

Caption:- Overview of guided storage layout configuration.

Task 10: Select the Installation Disk

The correct VMware virtual disk was selected and its size verified before proceeding.

```
FILE SYSTEM SUMMARY

MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /               10.000G   new ext4  new LVM logical volume ► ]
[ /boot          1.771G   new ext4  new partition of local disk ► ]

AVAILABLE DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   18.222G ► ]
free space                             8.222G ►

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]
```

File name: ubuntu_selected_disk.png

Caption:- Selecting the VMware virtual disk for installation.

```
USED DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   18.222G ► ]
ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ►

[ /dev/sda                             local disk                         20.000G ► ]
partition 1  new, BIOS grub spacer                        1.000M ►
partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ►
partition 3  new, PV of LVM volume group ubuntu-vg          18.225G ►
```

File name: ubuntu_disk_details.png

aption:- Verifying selected disk details and total size allocation.

Task 11: Review the Partitions

The proposed partition layout was reviewed and confirmed to belong to the correct virtual disk.

```
System load: 0.03      Processes:           217
Usage of /:  50.5% of 9.75GB Users logged in:       0
Memory usage: 21%      IPv4 address for ens33: 192.168.15.130
Swap usage:  0%
```

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.

To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

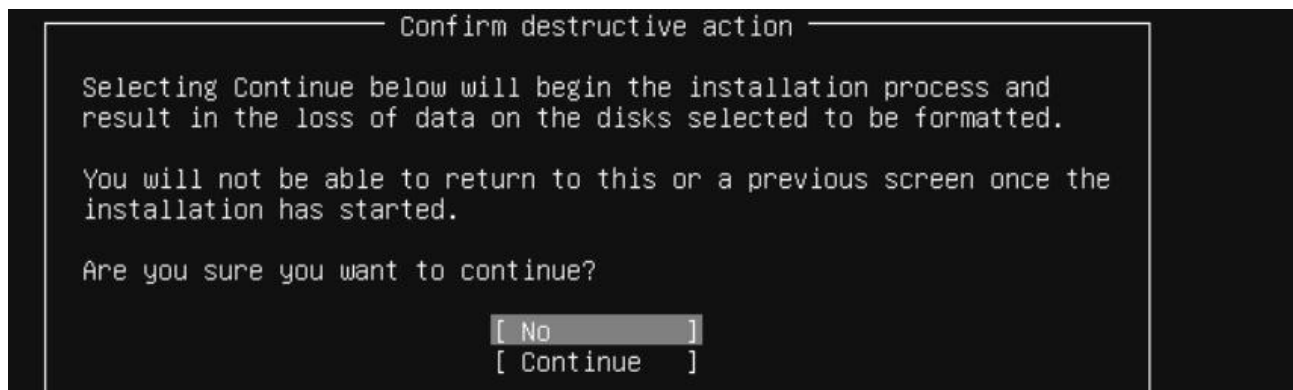
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

File name: ubuntu_partition_layout.png

Caption:- Reviewing proposed filesystem partition layout on `/dev/sda`.

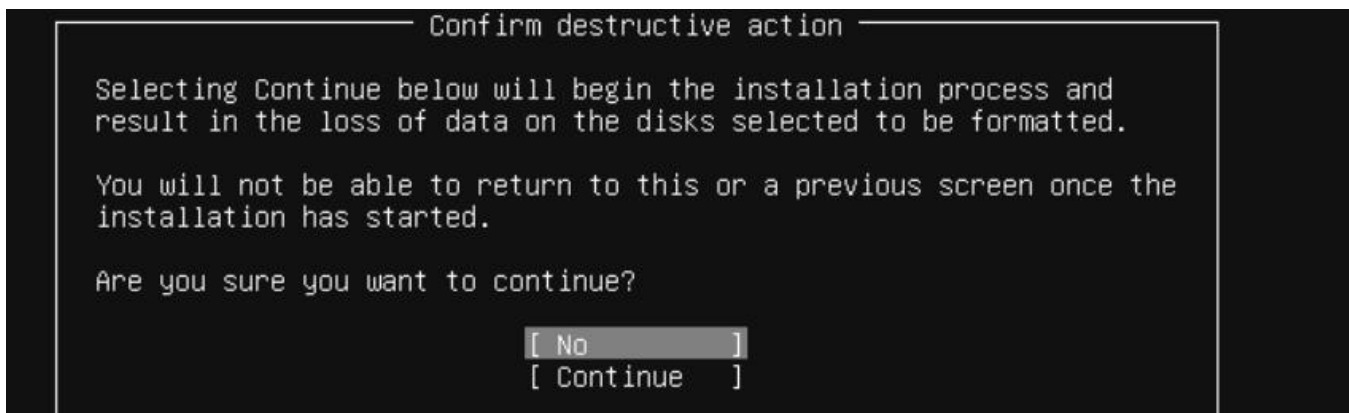
Task 12: Confirm the Storage Changes

The destructive-action warning was reviewed and storage changes were confirmed for the correct disk.



File name: ubuntu_storage_warning.png

Caption:- Destructive-action storage configuration warning prompt.

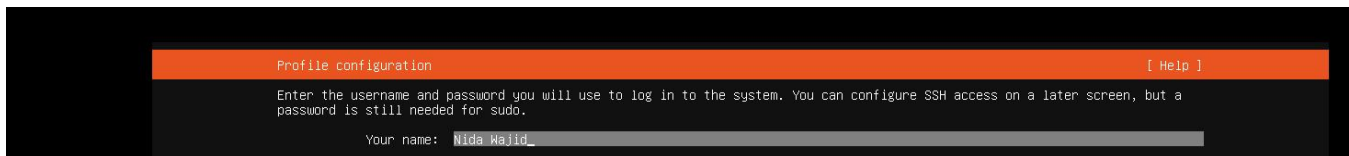


File name: ubuntu_storage_confirmed.png

Caption:- Confirming write changes to disk

Task 13: Configure the Ubuntu Profile

The Ubuntu user profile was configured with my actual name, server name, and GitHub-matching username.



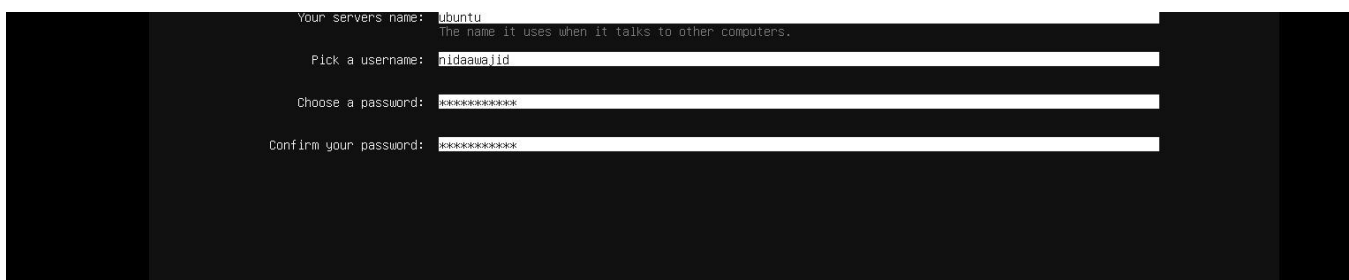
File name: ubuntu_actual_name.png

Caption:- Entering actual full name in user profile setup



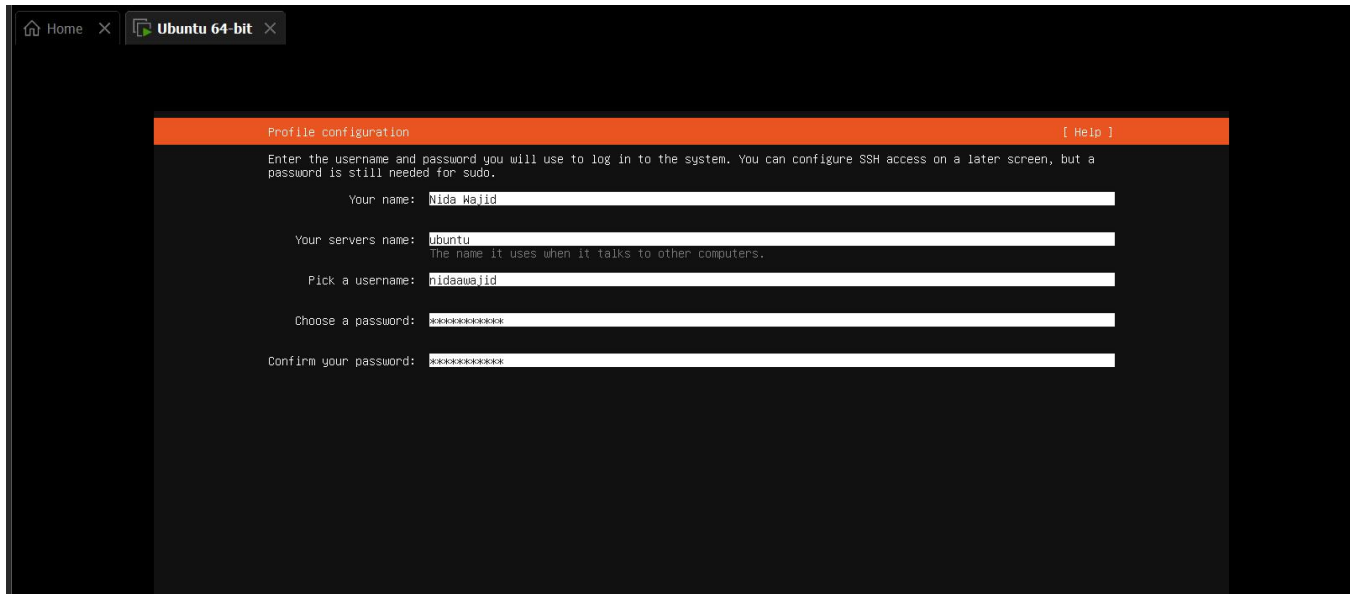
File name: ubuntu_server_name.png

Caption:- Configuring server hostname as ubuntu.



File name: ubuntu_username.png

Caption:- Setting Ubuntu username to match exact GitHub username (nidaawajid).

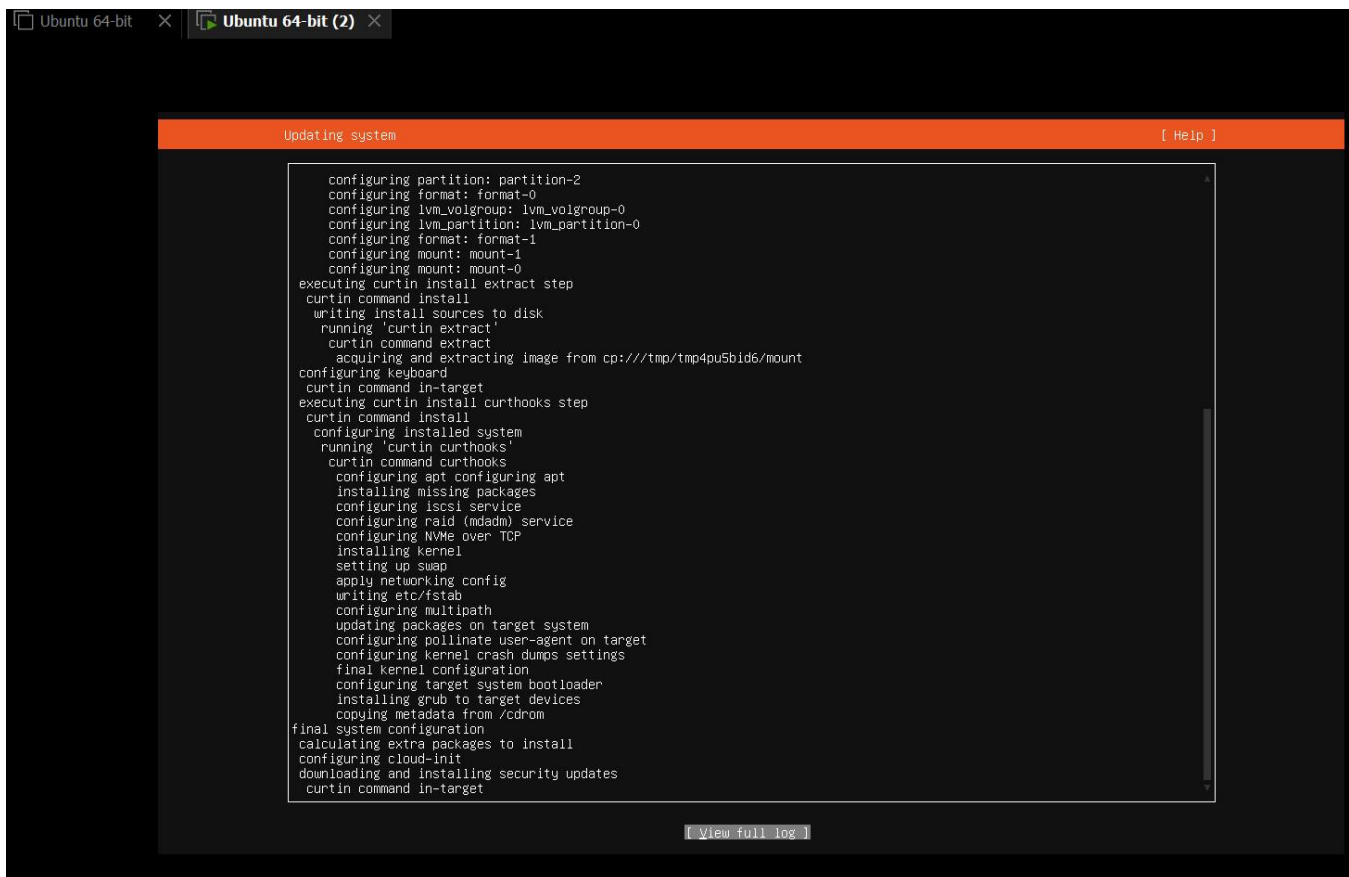


File name: ubuntu_profile_setup.png

Caption: Completed user account and profile configuration page.

Task 14: Complete the Software Installation

The Ubuntu Server software installation was completed without interruption.

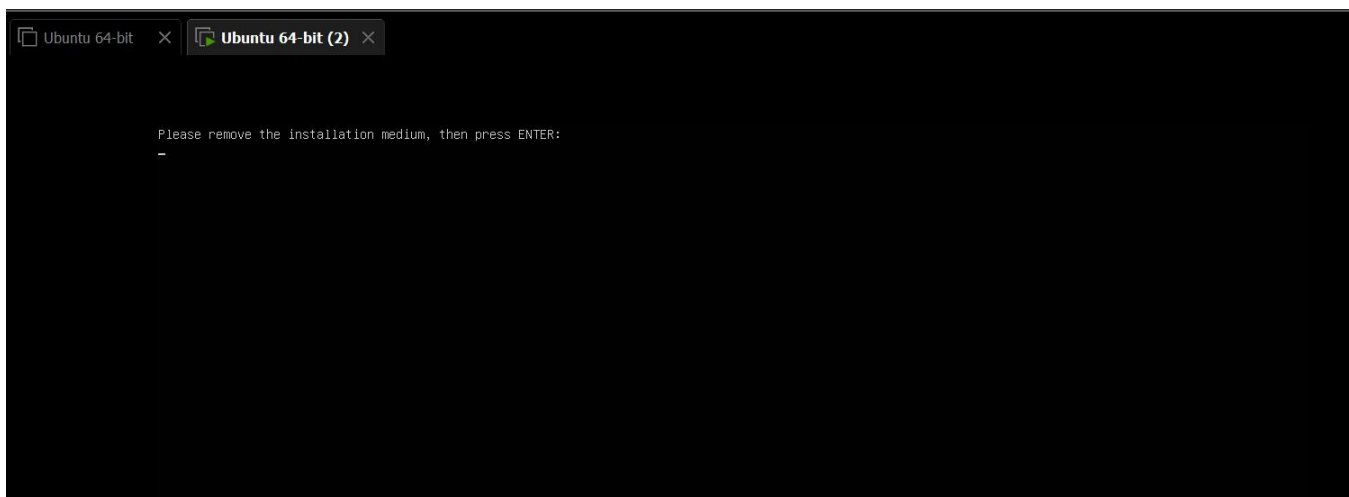


File name: ubuntu_installation_progress.png

Caption:- System packages and core installation progress.

Task 15: Reboot and Verify the Ubuntu Identity

The system was rebooted, and a successful terminal login confirmed the correct Ubuntu identity and prompt format.



File name: ubuntu_installation_complete.png

Caption:- Installation completed screen with reboot prompt

```
[ 20.238667] cloud-init[1155]: +----[SHA256]-----+
ubuntu login:
ubuntu login: nidaawajid
Password:
```

File name: ubuntu_login_screen.png

Caption:- Ubuntu Server local login prompt after initial reboot.

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
nidaawajid@ubuntu:~$ _
```

File name: ubuntu_terminal_login.png

Caption:- Successful terminal authentication

```
The programs included with the Ubuntu system are free software:
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
nidaawajid@ubuntu:~$ _
```

File name: ubuntu_identity_verified.png

Caption:- Terminal identity prompt verification (nidaawajid@ubuntu:~\$)

Task 16:-Find the Ubuntu Server IP Address:

```
nidaawajid@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:05:04:46 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnam enx000c29050446
    inet 192.168.6.131/24 metric 100 brd 192.168.6.255 scope global dynamic ens33
        valid_lft 1276sec preferred_lft 1276sec
    inet6 fe80::28c:29ff:fe05:446/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
nidaawajid@ubuntu:~$
```

File Name :ubuntu_ip_addr_command.png

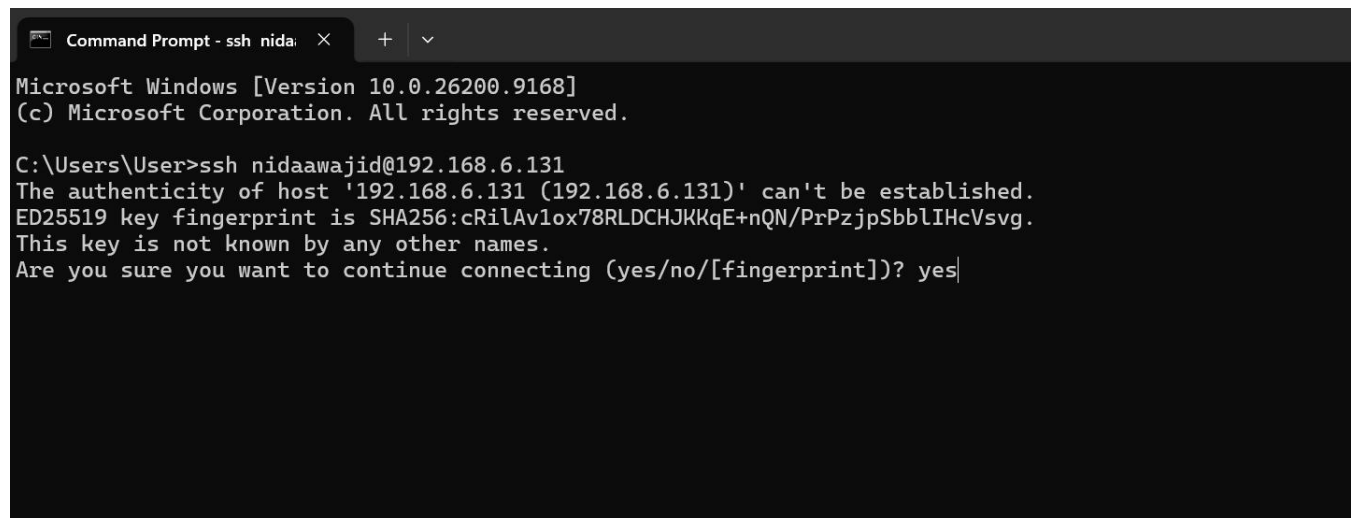
Caption:- Execution of ip addr command in Ubuntu terminal.

```
nidaawajid@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:05:04:46 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29050446
    inet 192.168.6.131/24 metric 100 brd 192.168.6.255 scope global dynamic ens33
        valid_lft 1276sec preferred_lft 1276sec
    inet6 fe80::20c:29ff:fe05:446/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
nidaawajid@ubuntu:~$
```

File Name :ubuntu_ip_address.png

Caption:- Displaying active IPv4 address (192.168.6.131).

Task 17:-Connect to Ubuntu Server from Windows:

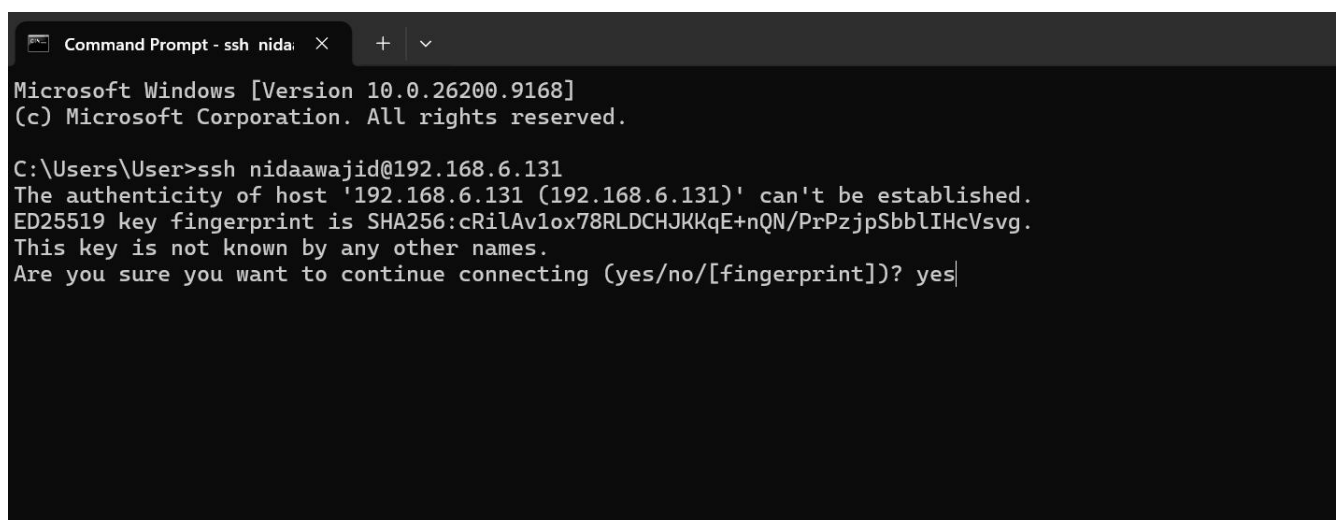


```
Command Prompt - ssh nida: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh nidaawajid@192.168.6.131
The authenticity of host '192.168.6.131 (192.168.6.131)' can't be established.
ED25519 key fingerprint is SHA256:cRiLAvIox78RLDCHJKKqE+nQN/PrPzjpSbblIHcVsvg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

File Name :windows_ssh_command.png

Caption:- Executing SSH command from Windows Command Prompt



```

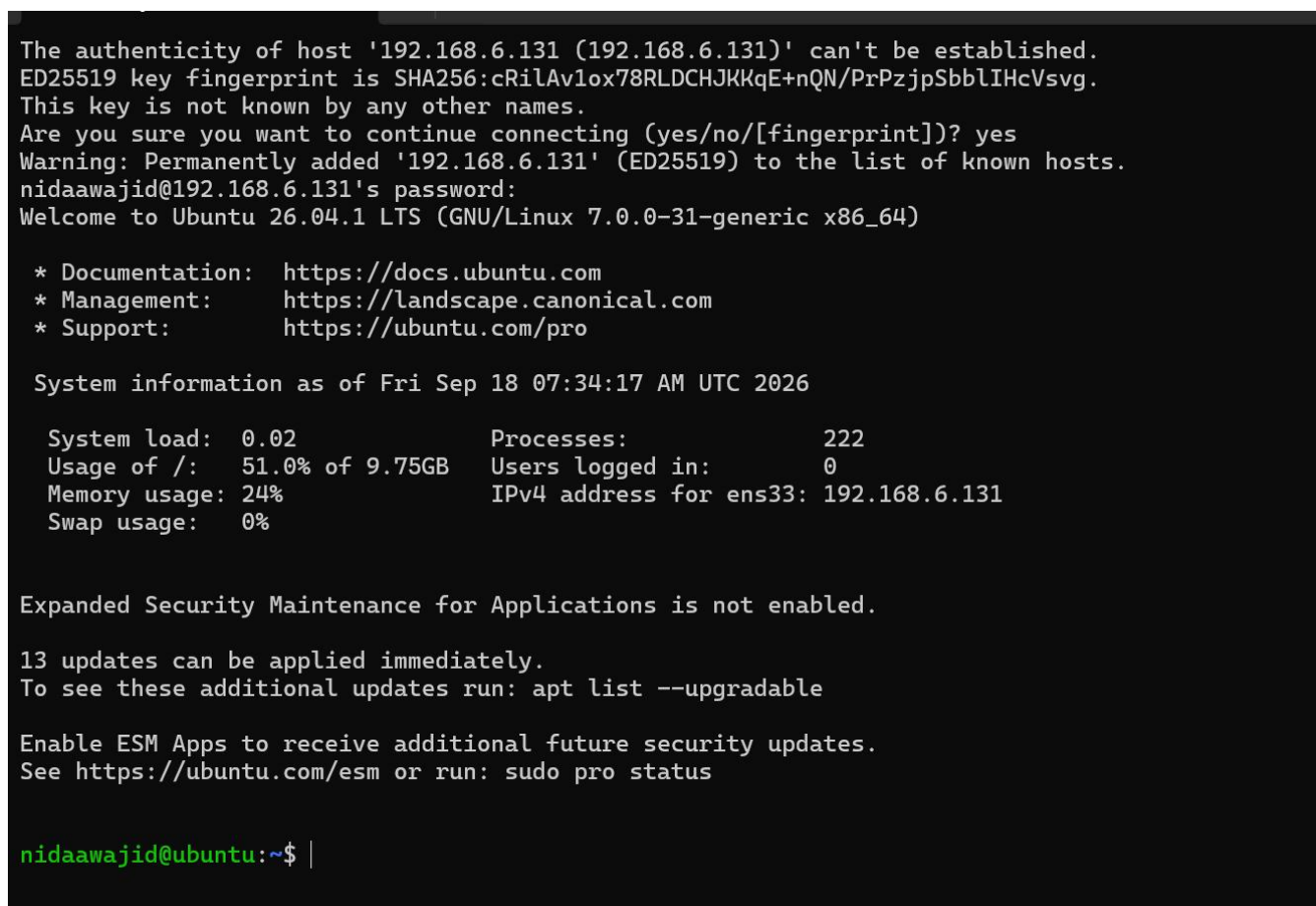
Command Prompt - ssh nida: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh nidaawajid@192.168.6.131
The authenticity of host '192.168.6.131 (192.168.6.131)' can't be established.
ED25519 key fingerprint is SHA256:cRiLAv1ox78RLDCHJkKqE+nQN/PrPzjpSbbLIHcVsvg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

```

File Name : windows_ssh_fingerprint.png

Caption:- Accepting RSA/ED25519 server fingerprint key



```

The authenticity of host '192.168.6.131 (192.168.6.131)' can't be established.
ED25519 key fingerprint is SHA256:cRiLAv1ox78RLDCHJkKqE+nQN/PrPzjpSbbLIHcVsvg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.6.131' (ED25519) to the list of known hosts.
nidaawajid@192.168.6.131's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 07:34:17 AM UTC 2026

System load:  0.02               Processes:            222
Usage of /:   51.0% of 9.75GB    Users logged in:     0
Memory usage: 24%               IPv4 address for ens33: 192.168.6.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

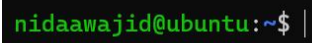
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

nidaawajid@ubuntu:~$ |

```

File Name: windows_ssh_login.png

Caption:- Successful remote SSH authentication session

A screenshot of a terminal window with a black background. The text 'nidaawajid@ubuntu:~\$ |' is displayed in a green monospace font. The prompt is followed by a vertical bar cursor.

```
nidaawajid@ubuntu:~$ |
```

File Name :windows_ssh_identity.png

Caption:- Remote SSH shell displaying required identity prompt (nidaawajid@ubuntu:~\$).